

to minimise the time and efforts that go into header processing. IPv6 headers are processed more efficiently at the intermediate routers, without having to parse through headers or recompute network-layer checksums, or even fragment and reassemble packets. Moreover, IPv6 extension headers can be expanded to accommodate whatever extension data is thought necessary for efficient IPv6 communication.

Better security for applications and networks. In IPv6, IPSec is a major protocol requirement and one of the factors in ensuring that IPv6 provides better security than IPv4. It contains a set of cryptographic protocols for ensuring secure data communication and key exchange. The main protocols used are authentication header protocol, encapsulating security payload protocol and Internet key exchange protocol.

End-to-end security mechanisms provide authentication and encryption abilities to all applications. Thus IPv6 eliminates the need for applications themselves to have integrated support for such abilities.

Better end-to-end connectivity. In the absence of unique IP addresses for each end, NAT creates difficulty in ensuring proper end-to-end services. IPv6 with its large address space no longer requires NAT and can ensure true end-to-end connectivity. This means peer-to-peer applications like VoIP and streaming media can work very effectively and efficiently with IPv6.

Better auto-configuration. IPv4 uses the stateful address auto-configuration protocol—dynamic host configuration protocol (DHCP)—to configure the node plugged into the network.

IPv6 too offers automatic configuration but, more importantly, simple configuration mechanisms, known as plug-and-play auto-configuration. It offers DHCPv6, which is an auto-configuration protocol similar to IPv4 DHCP that offers stateful address auto-configuration. IPv6 also offers stateless or serverless address auto-configuration, which enables a host to automatically configure its own IPv6 address.

A number of transition mechanisms such as dual IPv4/IPv6 stack, tunneling, dual IP stack using network address and port translation, proxy and translation techniques have been defined that allow IPv4 and IPv6 to coexist till the time complete migration to IPv6 is not feasible

Better QoS/prioritised delivery. Quality of Service (QoS) is given a special boost in the IPv6 protocol. The IPv6 header contains a new field called 'flow label,' which defines how packets are identified and handled quickly and efficiently by the routers. This ensures a high degree of QoS especially for peer-to-peer applications like VoIP and other real-time applications.

Better mobility features. With IPv6, mobility support using Mobile IPv6 (MIPv6) has been made mandatory. Route optimisation is a built-in feature for Mobile IPv6. Further, features like neighbour discovery and address auto-configuration allow mobile nodes to function in any location without requiring the services of any special router. MIPv6 can be used to achieve seamless mobility by allowing handovers between different access technologies. There is no ingress-filtering problem in Mobile IPv6 because the correspondent node uses care-of-address (address of present location) as the source address.

Better multicast and anycast abilities. IPv6 extends the multicasting capabilities of IPv4 by offering a large multicast address range. Obviously, this limits the degree to which the information packets have to be propagated, and significantly improves the network efficiency. Multicasting requires the presence of fewer packets in the network, thereby optimising bandwidth utilisation and also reducing the resources required at each

network node.

IPv6 also improves dramatically on the concept of anycast services, which are available in a very minimal form in IPv4.

Ease of administration. IPv6 provides automatic network renumbering, which enables smoother switchovers or mergers. Another useful administrative feature of IPv6 is its multihoming technique, using which simultaneous connections are established to two Internet service providers (ISPs). When service to one ISP is lost, a back-up connection to the Internet ensures greater reliability of services.

Efficient, hierarchical addressing and routing infrastructure. IPv6 global addresses used on the IPv6 portion of the Internet are designed to create an efficient, hierarchical and summarisable routing infrastructure that is based on the common occurrence of multiple levels of Internet service providers.

Smooth transition from IPv4

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The IPv6 format will allow countless network addresses and almost eliminate the need for NAT. Nevertheless, the number of IP addresses should not be considered as the only reason to move to the new protocol. Other characteristics like greater network scalability, security, data integrity, QoS, auto configuration and better network routing needed for new business needs have also been integrated in IPv6, which thus holds the promise of achieving end-to-end security, mobile communications, quality of service and simplified system management. ●

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